

Writing Inequalities from Verbal Constraints

Answer Key

Grade 7

Quick Answers

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|--|--|
| 1. $(-\infty, 18]$ | 6. $[37, \infty)$ |
| 2. $(12, \infty)$ | 7. $(-\infty, 28]$ |
| 3. $[9, \infty)$ | 8. $(-\infty, 4)$ |
| 4. (b) the solution = $(-\infty, 7]$, — | 9. (b) the solution = $(-\infty, 25)$, — |
| 5. $(-\infty, 14)$ | 10. (b) the solution = $(-\infty, 15]$, — |
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What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 7

7.EE.B.4 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 14 tagged checks.

1. Lift limit 450 kg, boxes of 25 kg. Inequality for b and its solution.

Solution: “No more than” allows the limit itself, so the sign is $\leq$: $25b \leq 450$. Dividing both sides by 25 keeps the direction because 25 is positive: $b \leq 18$. The test table agrees — 18 boxes weigh exactly 450 kg and are allowed, 19 boxes weigh 475 kg and are not. $b \leq 18$

2. A film for people over 12. Inequality for the age y .

Solution: “Over 12” does not include 12 itself, so the sign is strict: $y > 12$, which is already solved. A twelve-year-old may not watch. $y > 12$

3. Pass with at least 36 points, 4 points a question. Inequality for q .

Solution: “At least” includes the boundary, so $4q \geq 36$. Divide both sides by 4: $q \geq 9$. Nine correct questions score exactly 36 and pass. $q \geq 9$

4. Taxi: 4 dollars to start plus 3 dollars per km, at most 25 dollars.

Solution: (a) **Instructor-judged.** The distance charge is $3k$ and the start charge is paid once, and “at most” allows spending the whole 25 dollars: $3k + 4 \leq 25$. Accept $4 + 3k \leq 25$ or $25 \geq 3k + 4$. A strict $<$ is wrong, and so is an equation with an equals sign.

(b) Subtract 4 from both sides: $3k \leq 21$. Divide by 3: $k \leq 7$ kilometres.

$$k \leq 7$$

5. Parcel under 20 kg, 6 kg already packed. Inequality for m .

Solution: “Less than” excludes the boundary, so $m + 6 < 20$. Subtract 6 from both sides: $m < 14$ kilograms. Exactly 14 kilograms of contents would make the parcel weigh 20 kg, which is not allowed.

$$m < 14$$

6. Badge needs at least 60 km walked; 23 km already done.

Solution: “At least” includes the boundary: $d + 23 \geq 60$. Subtract 23: $d \geq 37$ kilometres. The table agrees — 37 further kilometres reaches exactly 60 and earns the badge, 36 does not.

$$d \geq 37$$

7. Shelf holds at most 45 kg; bookends 3 kg, each book $\frac{3}{2}$ kg.

Solution: The bookends are on the shelf once and each book repeats, so $\frac{3}{2}n + 3 \leq 45$. Subtract 3: $\frac{3}{2}n \leq 42$. Multiply both sides by $\frac{2}{3}$: $n \leq 28$ books.

$$n \leq 28$$

8. Fewer than 24 players in 6 equal teams. Inequality for p .

Solution: Six teams of p players is $6p$ players in all, and “fewer than” is strict: $6p < 24$. Divide both sides by 6: $p < 4$ players on a team.

$$p < 4$$

9. Pool under 900 litres; 150 already in, tap adds 30 litres a minute.

Solution: (a) **Instructor-judged.** The tap contributes $30t$ litres and the 150 litres are already there, and “fewer than” is strict: $30t + 150 < 900$. Accept $150 + 30t < 900$. A non-strict $\leq$ is wrong, and $30(t + 150)$ is wrong — the 150 litres are not added every minute.

(b) Subtract 150: $30t < 750$. Divide by 30: $t < 25$ minutes.

$$t < 25$$

10. Bus tour: 30 dollars for the driver plus 12 dollars a person, at most 210 dollars.

Solution: (a) **Instructor-judged.** The per-person charge is $12n$ and the driver’s 30 dollars is paid once, and “at most” allows spending all 210 dollars: $12n + 30 \leq 210$. Accept $30 + 12n \leq 210$. A strict $<$ is wrong.

(b) Subtract 30: $12n \leq 180$. Divide by 12: $n \leq 15$ people.

$$n \leq 15$$

(c) **Instructor-judged.** The solution set contains every real number up to and including

15 — including 7.4 and -3 — but a group size must be a whole number and cannot be negative, so only the whole numbers from 0 (or 1) up to 15 are actually possible. Full credit needs the whole-number point and the observation that the solution set on its own does not carry that restriction; the situation does.
